

词汇: *astronomy* 天文学

- 1 You may have heard of the idiom 'once in a blue moon' to describe an event that doesn't happen often. But can the Moon really be blue? Our closest neighbour in this **vast galaxy**, the Moon, **orbits** the Earth like a natural **satellite**. Unlike many of the other planets in our **Solar System**, ours only has the one moon. If you flew to it, you would land on grey rocks. This is because the **lunar** surface is made from mostly magnesium, iron and other minerals of that colour. But, if you're lucky to have clear skies under the curtain of night, you may be able to admire its **luminosity**. You might even see that it seems to change shape and **hue**.
- 2 It's important to understand that the Moon **reflects** light from the Sun as it orbits the Earth, and that light bounces back to the Earth. The position of the Moon, Sun and the Earth determines how much of the Moon we see. The new Moon appears almost **invisible** in the sky as it is in total darkness. This is the opposite to the full Moon which is completely **lit up**. There are eight different **phases** to the lunar **cycle** between a full Moon and a new Moon, each lasting around 28 days. In fact, the word 'month' takes its **root** from 'Moon' because it was almost equal to the lunar phases.
- 3 The other phases include the crescent, quarter and gibbous phase. Crescent originates from the Latin verb 'crescere', meaning 'to increase', and so it is the phase just before and after a new Moon as it takes a **concave** shape. The quarter will show half of the Moon. 'Gibbous' gets its root from the Latin term 'gibbus' which applied to the Moon's **convex** shape. These shapes are determined by the adjectives 'waning', to get smaller, and 'waxing', to get bigger.
- 4 So now you know how we've named the shapes, what about those **mesmerising** colours? The **light waves** that reach our eyes can vary due to the Moon's position on the **horizon** and pollution. This is known as Rayleigh **scattering**. When the Moon is low, light travels through the **densest** part of the atmosphere and shorter waves of blue scatter away. This leaves only the red and orange light waves to reach us. A warmer atmosphere, as well as smoke from wildfire, may increase the scattering effect as well, making the Moon appear less **dull** in spring and summer.

This is why it's more likely for you to look up and see a pink Moon in April, a golden Moon in May and July, and even a red coloured Moon in the month of June.

测验与练习

一、将标题和段落配对。

Paragraph 1 _____

Paragraph 2 _____

Paragraph 3 _____

Paragraph 4 _____

- a. How do we see the Moon?
- b. The different colours of the Moon
- c. The different shapes of the Moon
- d. Our Moon
- e. Things we can find on the Moon

二、根据文章内容选出正确的选项。

1. We are the only planet in the Solar System with one Moon.

- a. True
- b. False
- c. Not given

2. The Moon is grey because...

- a. the light reflected from the Sun is dull.
- b. the rocks are made of grey minerals.
- c. we see it through grey clouds.

3. What is the opposite of a new Moon?

- a. a dark sky with an invisible Moon
- b. a bright Moon, only half full in shape
- c. a completely lit up Moon, full in shape

4. Which of these adjectives describes something that is growing?

- a. waning
- b. waving
- c. waxing

5. We see the most mesmerising colours of the Moon when it is high in the sky.

- a. True
- b. False
- c. Not given

三、用方框中的单词完成对本文的总结。

It does feel rather **1)** _____ to look out onto the **2)** _____ and see a brightly coloured Moon. The **3)** _____ of the Moon is determined by how low or high it is in the sky and the **4)** _____ of light waves in our atmosphere. Whereas the shape is determined by where the Moon as it **5)** _____ our Earth.

dull	horizon	hue
mesmerising		orbits
scattering		Solar System

（答案见词汇表后）

词汇表

vast	巨大的
galaxy	星系
orbit	沿轨道围绕...运行
satellite	卫星
Solar System	太阳系
lunar	月亮的
luminosity	光度，亮度
hue	色调，颜色
reflect	反射
invisible	看不见的
light up	照亮
phase	阶段
cycle	周期
root	词根
concave	凹的，凹面的
convex	凸的，凸出的
mesmerising	令人着迷的
light wave	光波
horizon	地平线
scattering	散射
densest	密度最大的，最浓密的
dull	暗淡的

答案

一、将标题和段落配对。

Paragraph 1 **d) Our Moon**

Paragraph 2 **a) How do we see the Moon?**

Paragraph 3 **c) The different shapes of the Moon**

Paragraph 4 **b) The different colours of the Moon**

二、根据文章内容选出正确的选项。

1. **a. True.** *Unlike many of the other planets in our Solar System, **ours only has the one Moon.***

2. **b. ...** *This is because **the lunar surface is made from mostly magnesium, iron and other minerals of that colour.***

3. **c. ...** *The new Moon appears almost invisible in the sky as it is in total darkness. This is the opposite to **the full Moon which is completely lit up.***

4. **c.** *These shapes are determined by the adjectives "waning", to get smaller, and "**waxing**", to get bigger.*

5. **b. False.** *When the Moon is low, light travels through the densest part of the atmosphere and shorter waves of blue scatter away. **This leaves only the red and orange light waves to reach us.** The mesmerising colours do not appear when the Moon is high.*

三、用方框中的单词完成对本文的总结。

It does feel rather **mesmerising** to look out onto the **horizon** and see a brightly coloured Moon. The **hue** of the Moon is determined by how low or high it is in the sky and the **scattering** of light waves in our atmosphere. Whereas the shape is determined by where the Moon is as it **orbits** our Earth.